

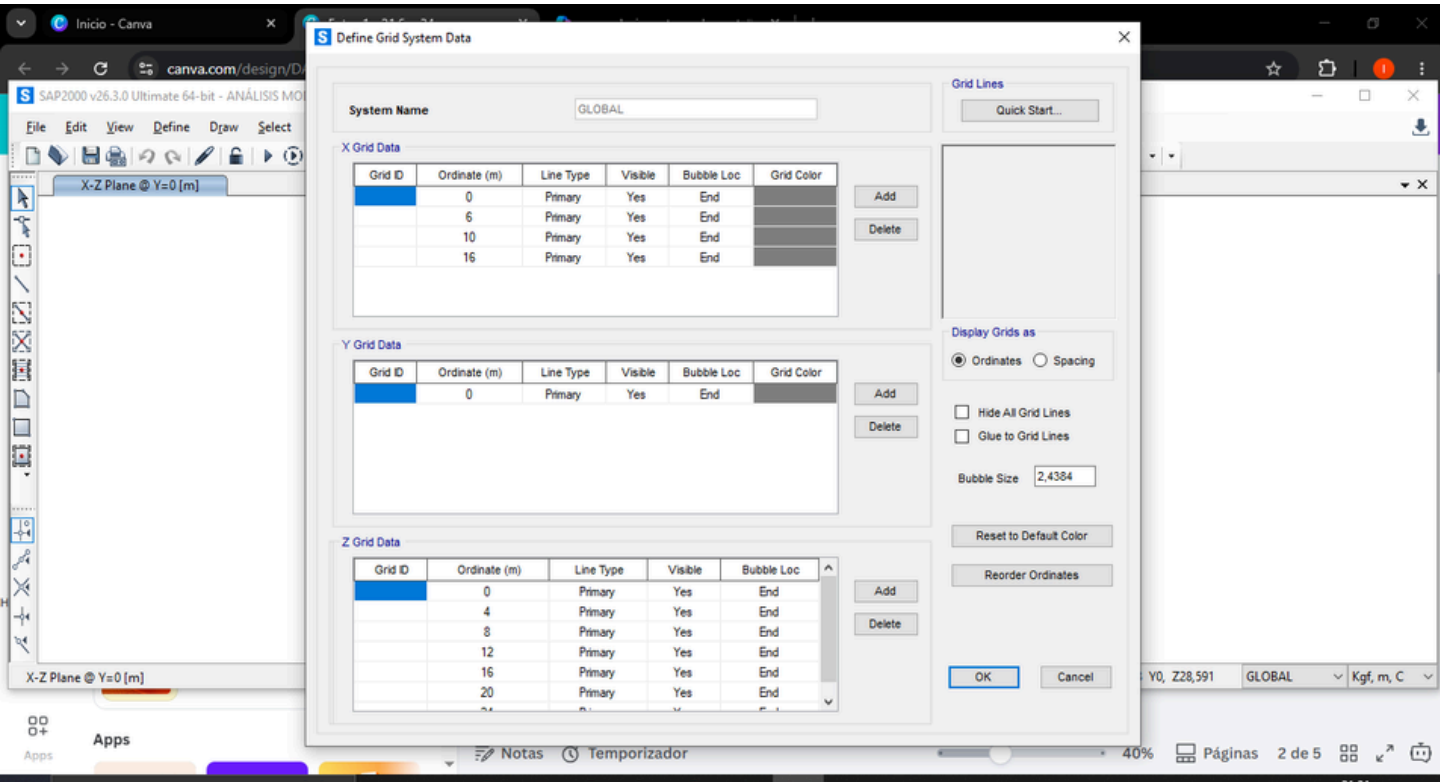
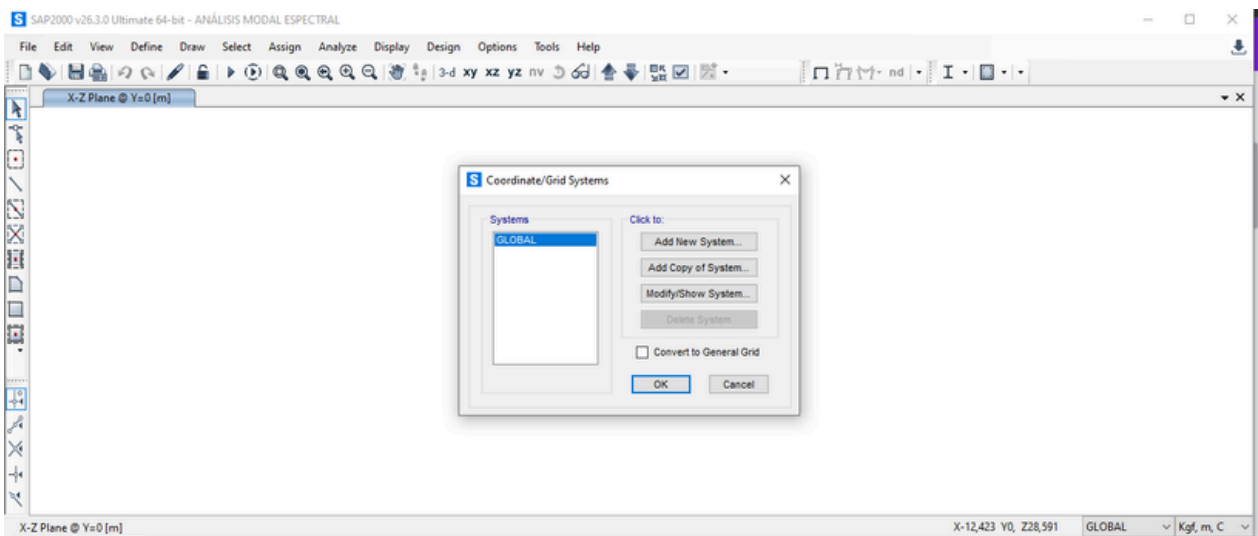
**UNIVERSIDAD SAN CARLOS DE GUATEMALA
DEPARTAMENTO DE ESTUDIO SUPERIO POSTGRADO
ANALISIS ESTRUCTURAL AVANZADO LINEAL Y NO LINEAL
ING. DANIEL ANTONIO QUINTANA ARCHILA**

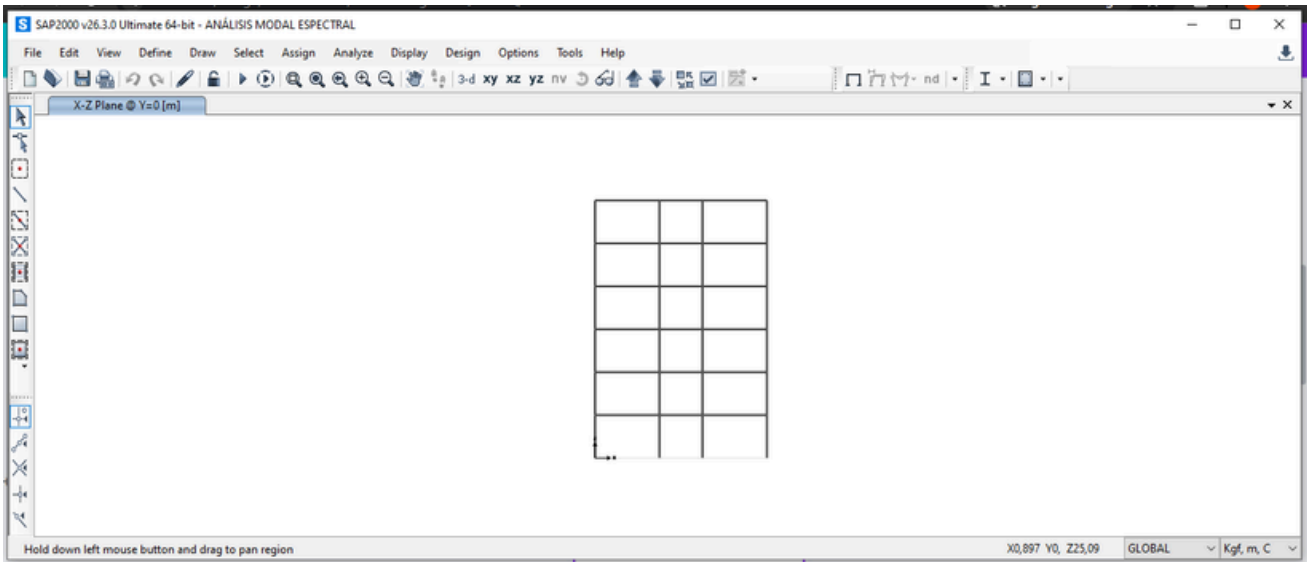


Capturas de pantalla del software de análisis estructural "SAP2000"

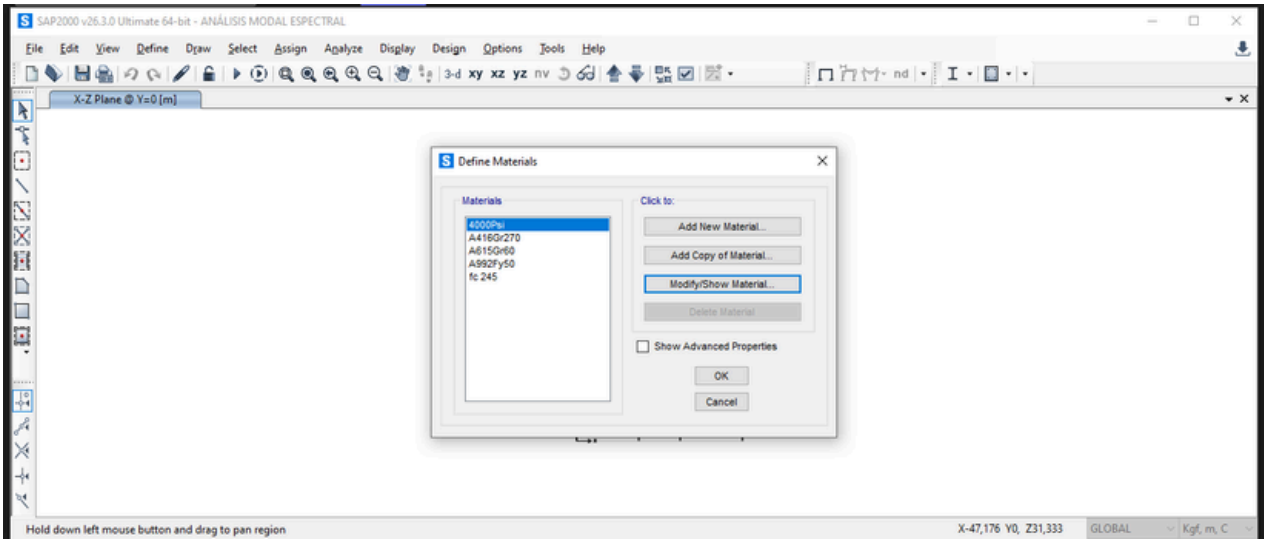
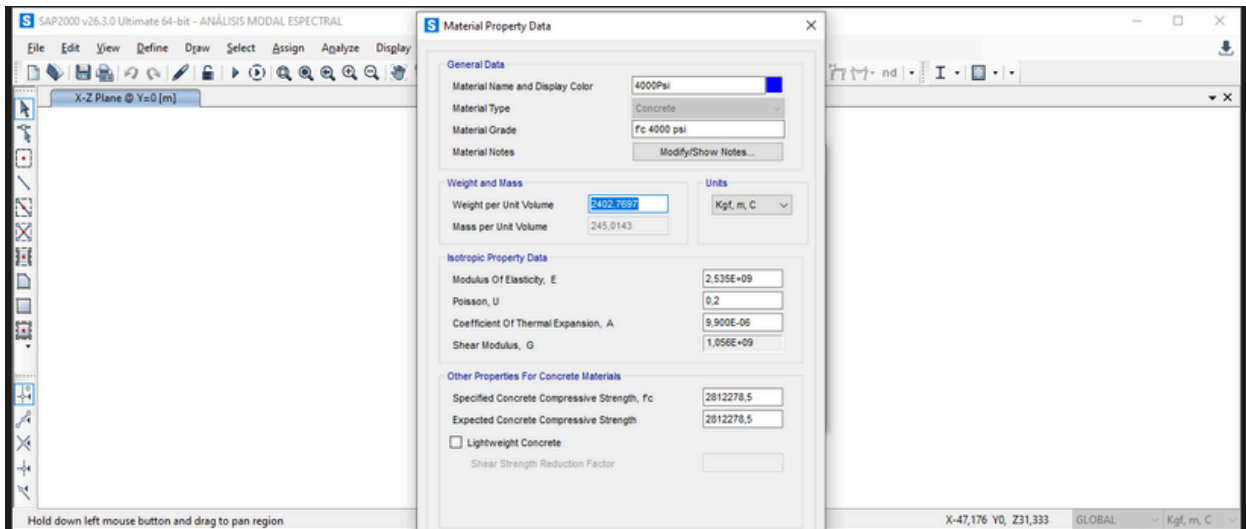
**MARLON IVAN CARRETO RIVERA
CARNET 201230088**

Agregando las coordenadas del modelo

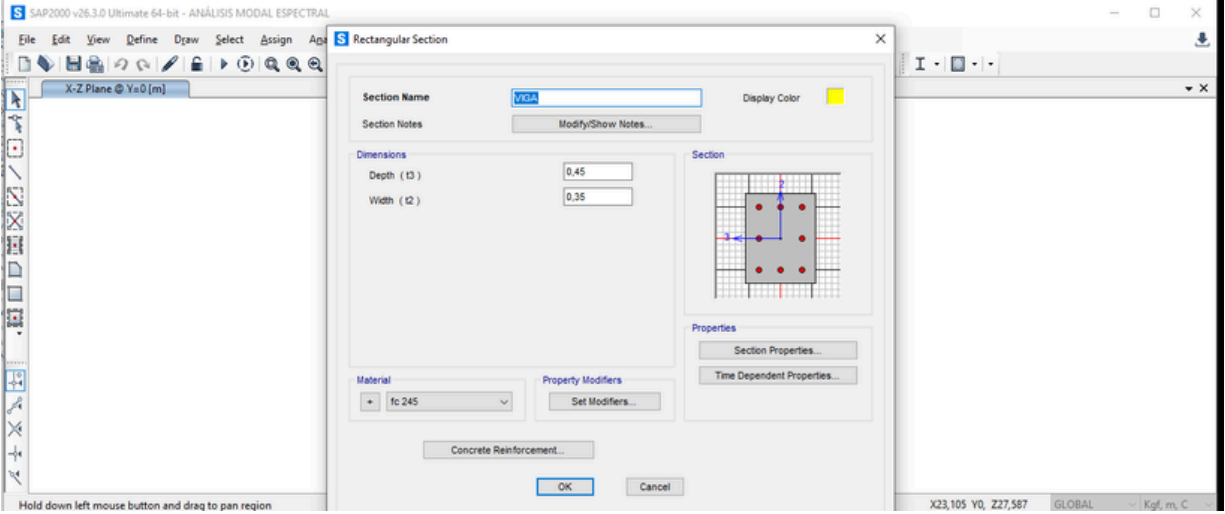
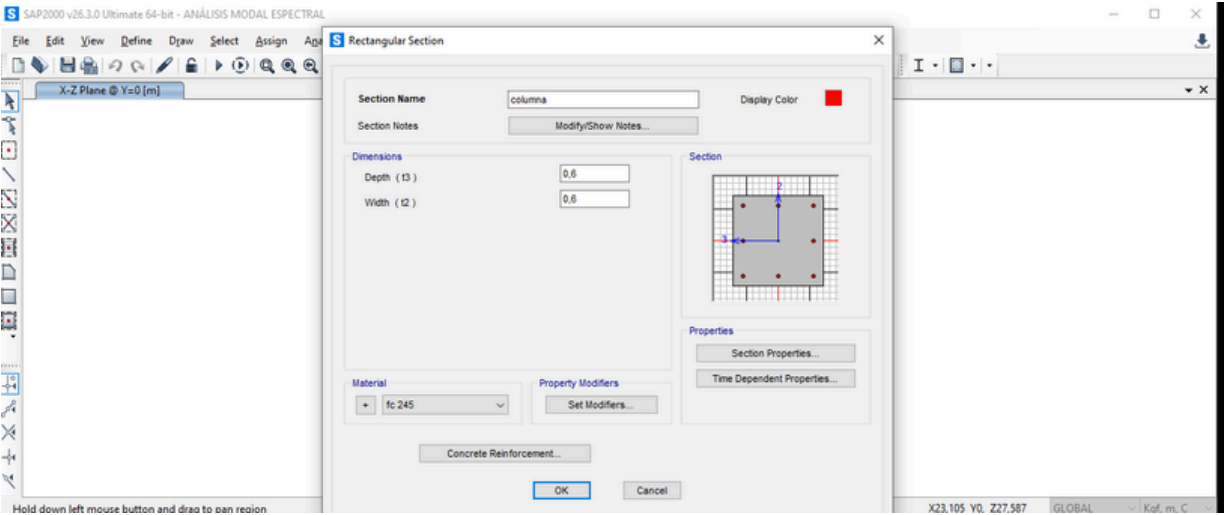
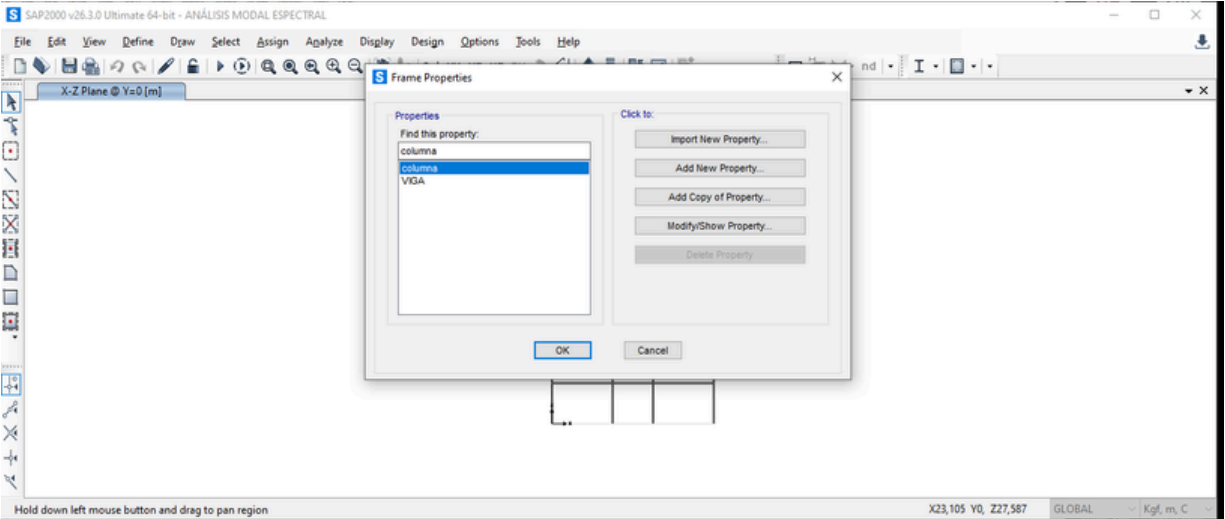




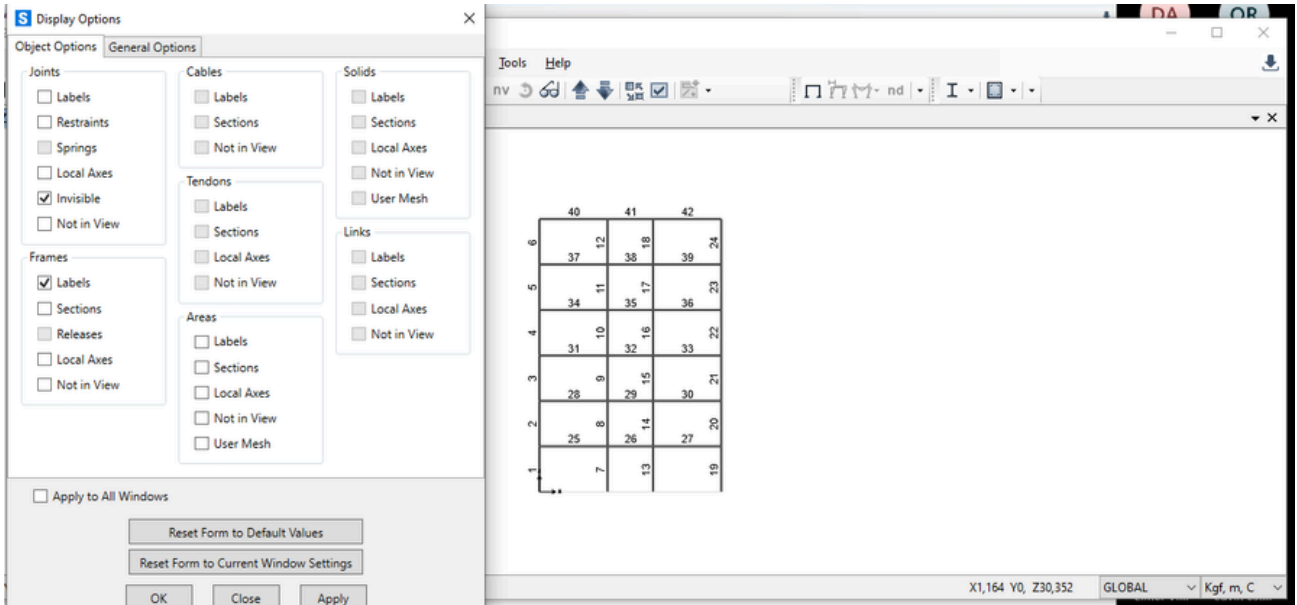
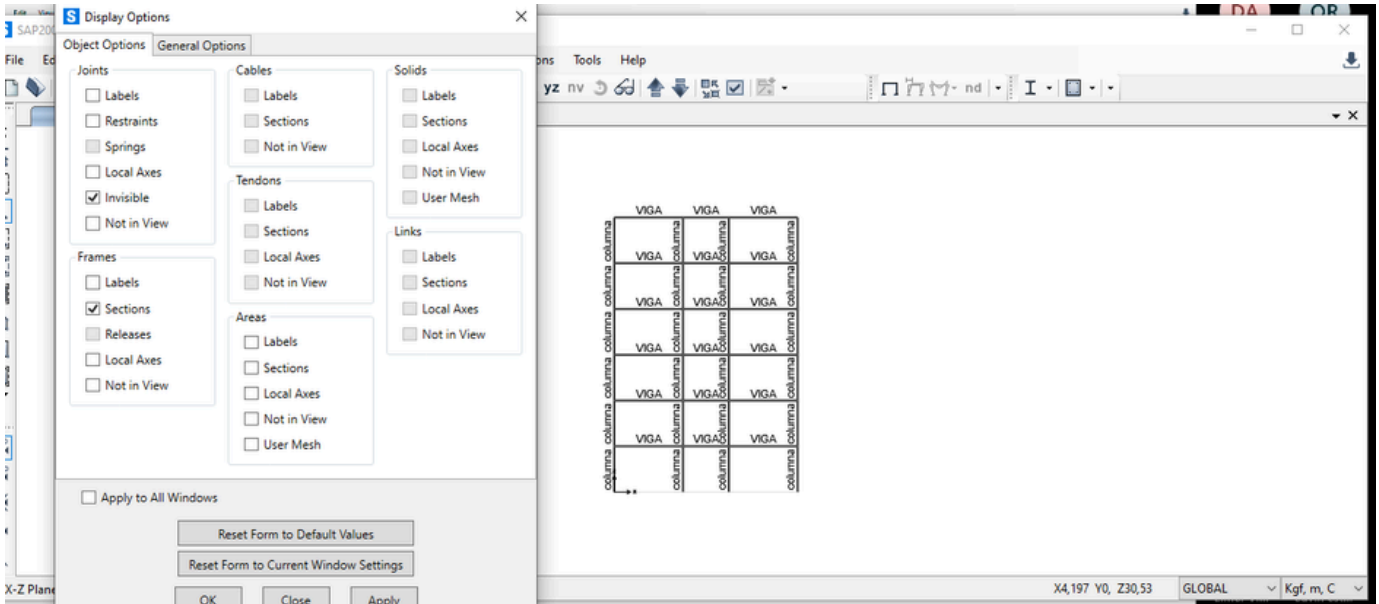
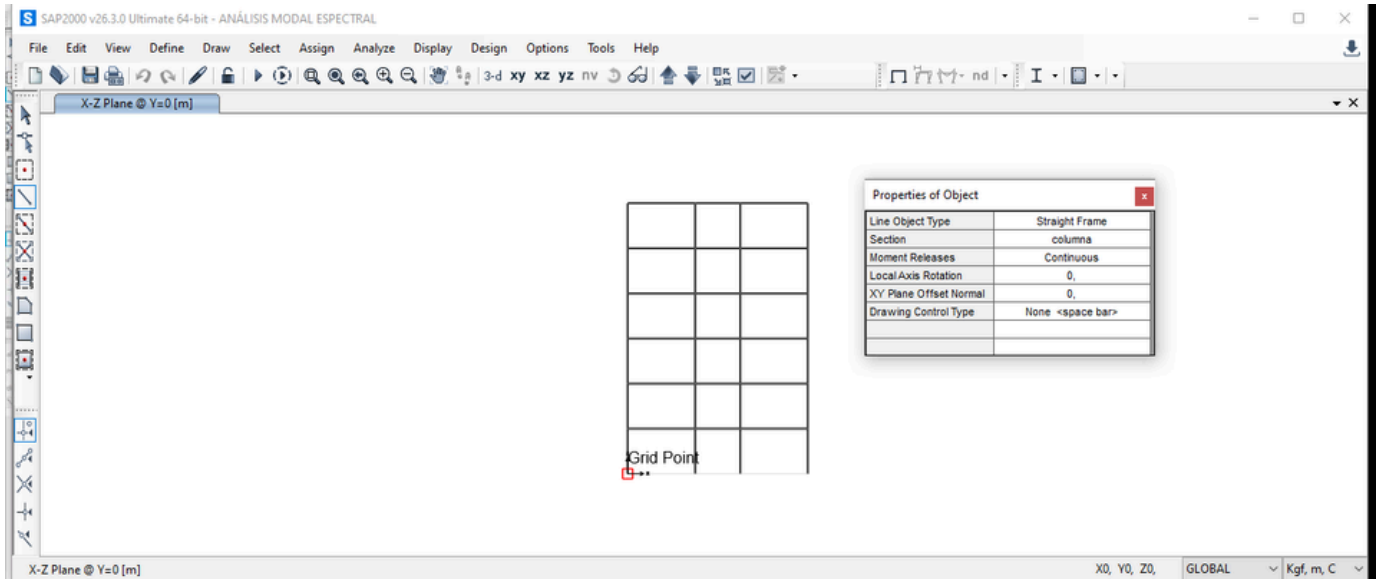
AGREGANDO TIPOS DE MATERIALES



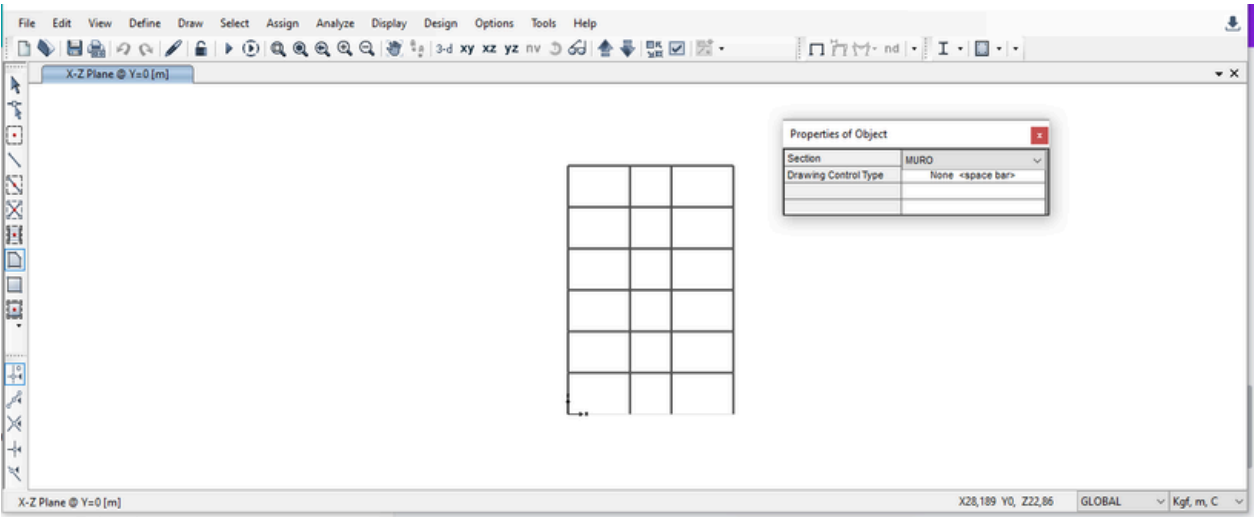
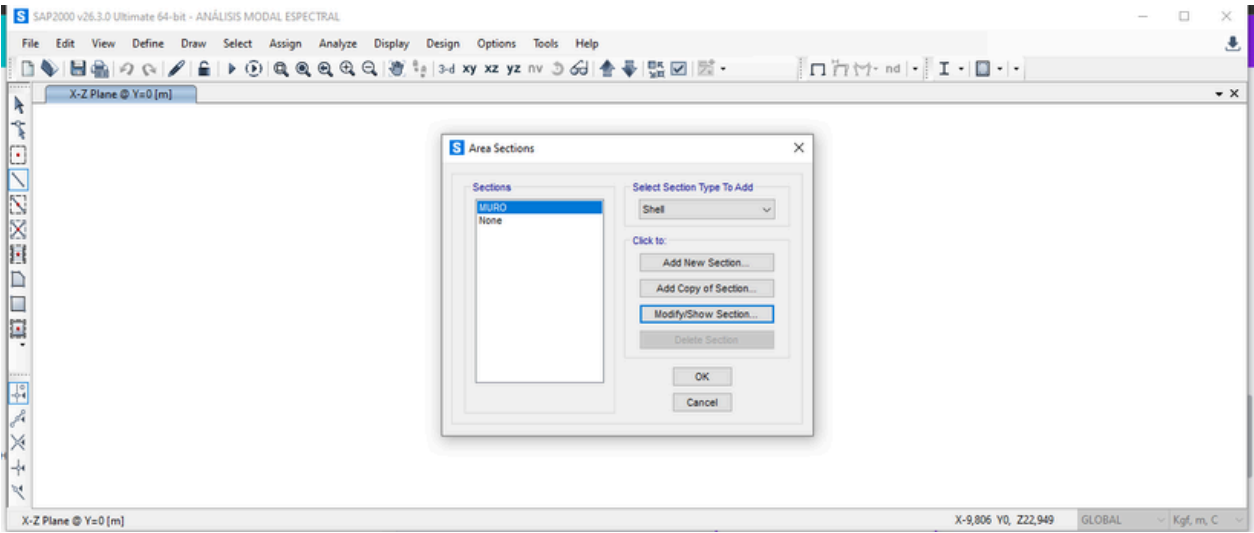
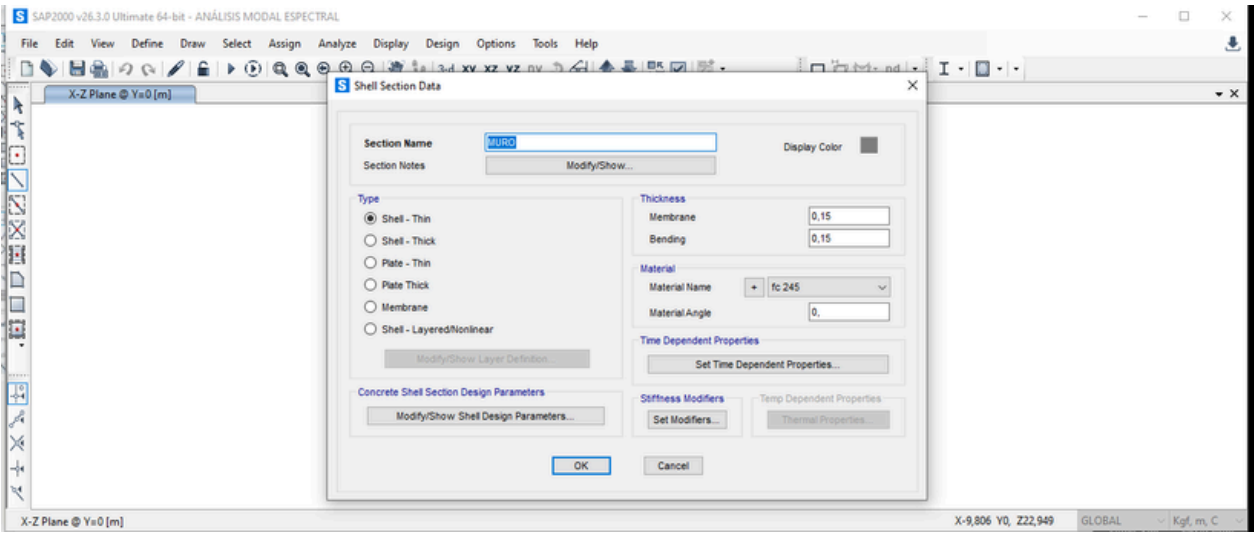
DEFINIENDO PROPIEDADES DE SECCIÓN, VIGA Y COLUMNA

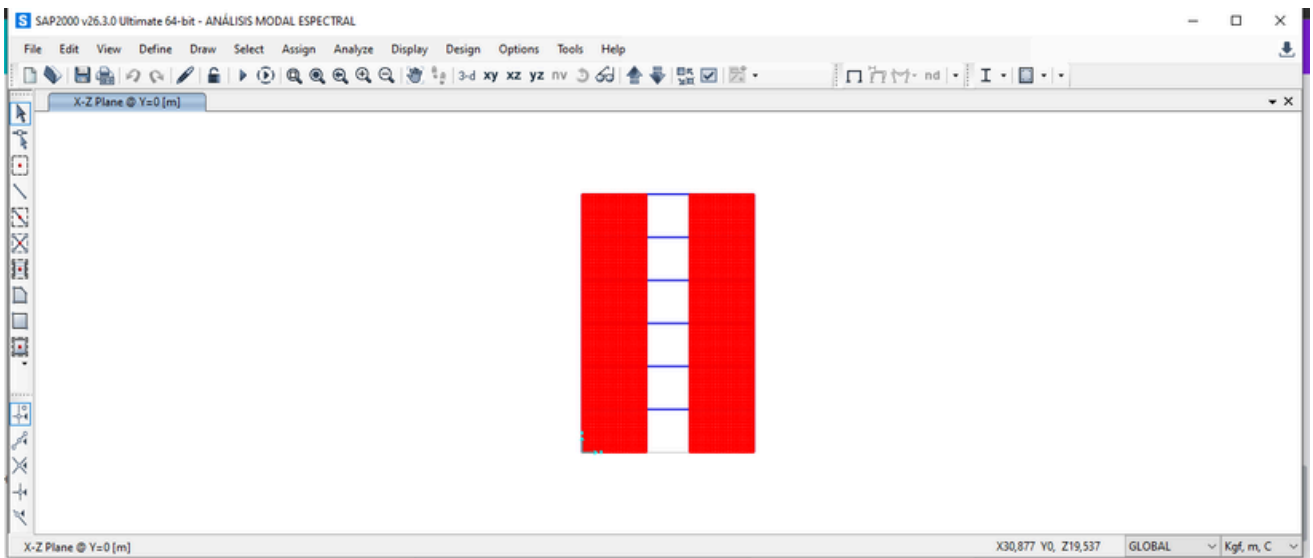
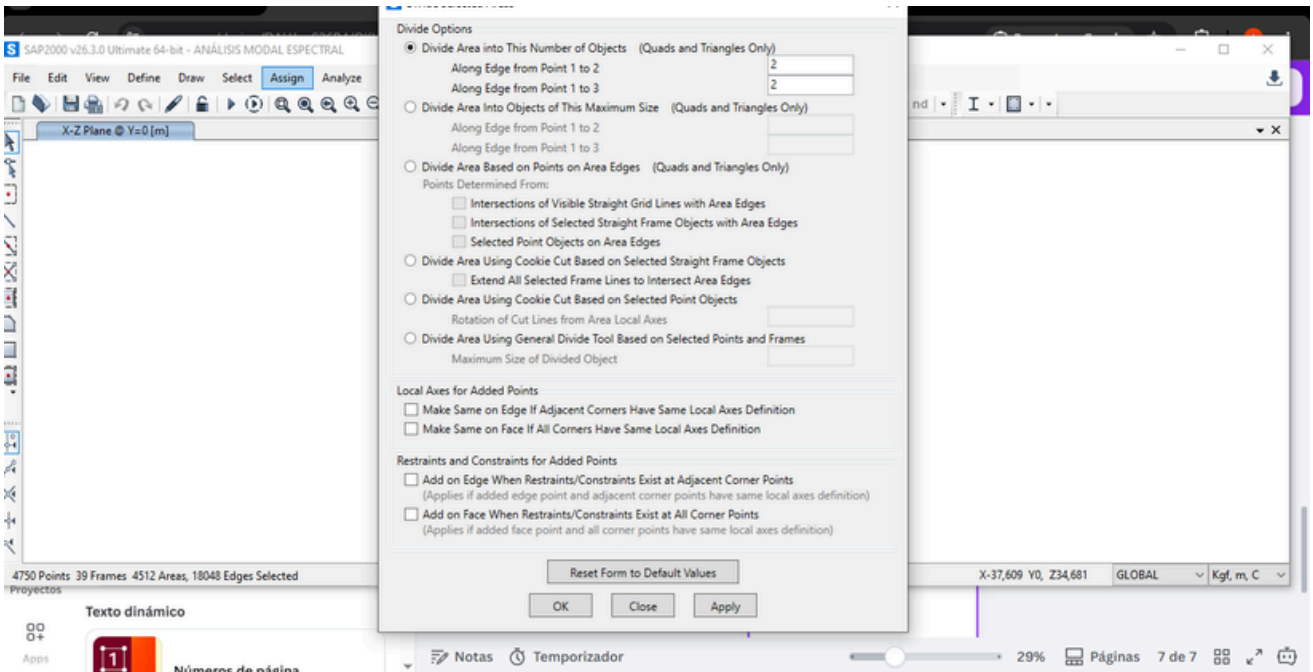
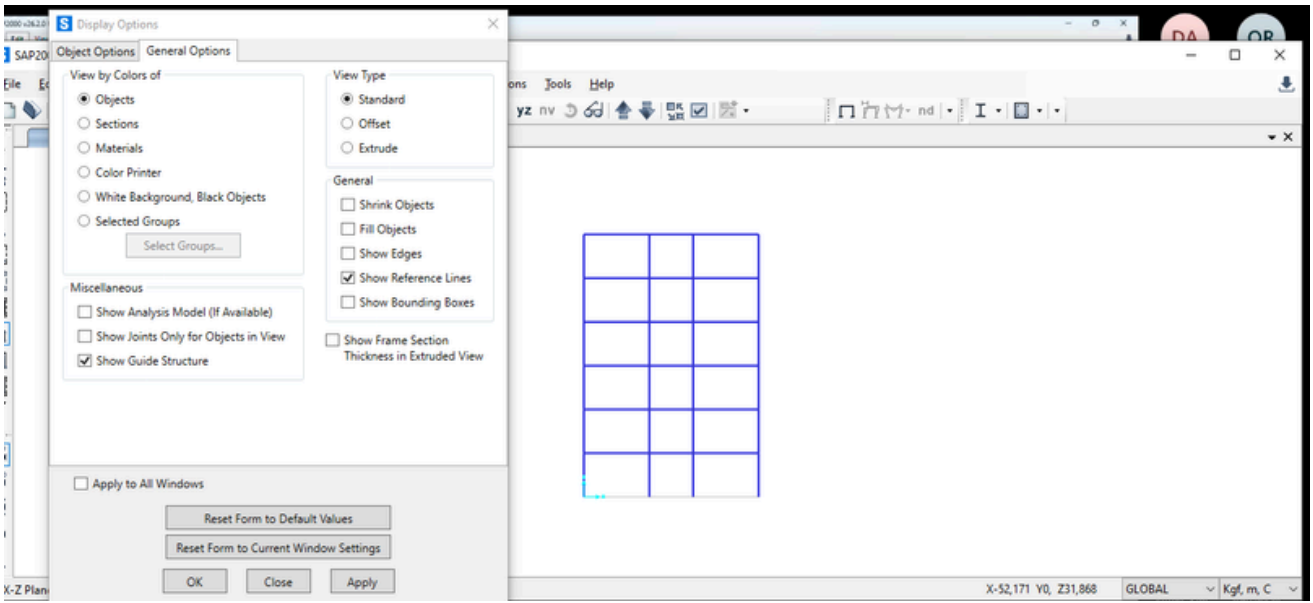


ESTABLECIENDO VIGAS Y COLUMNAS EN LA ESTRUCTURA

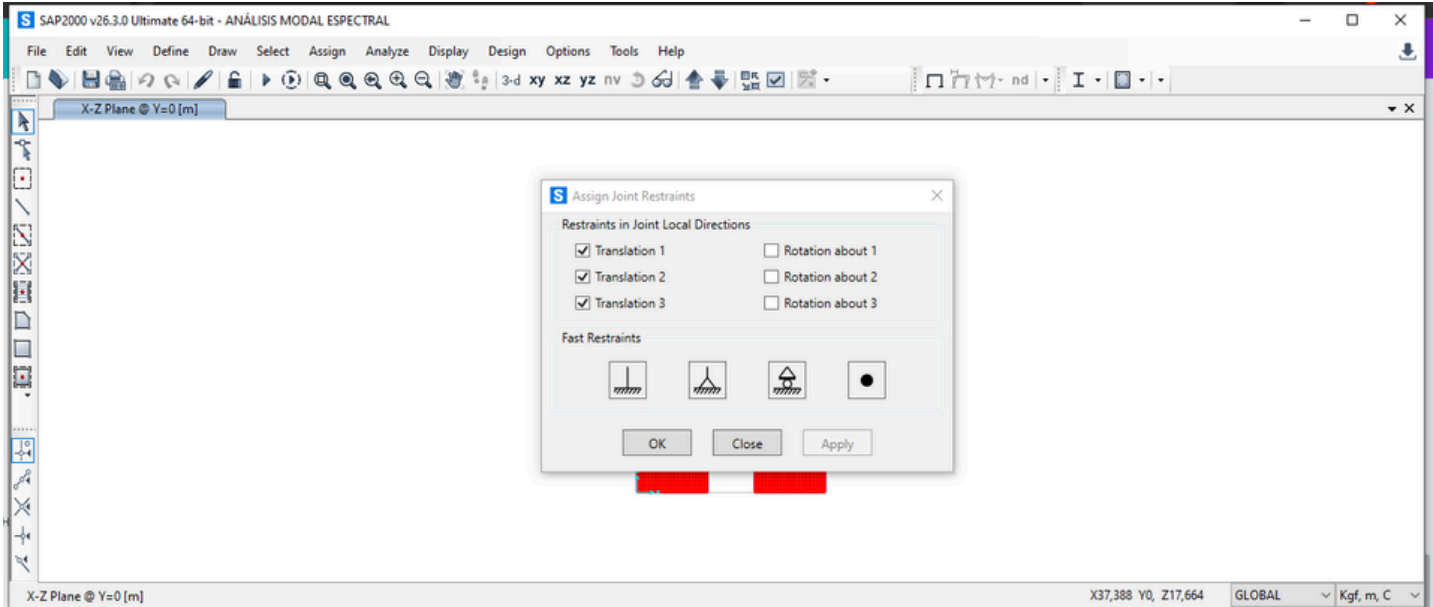


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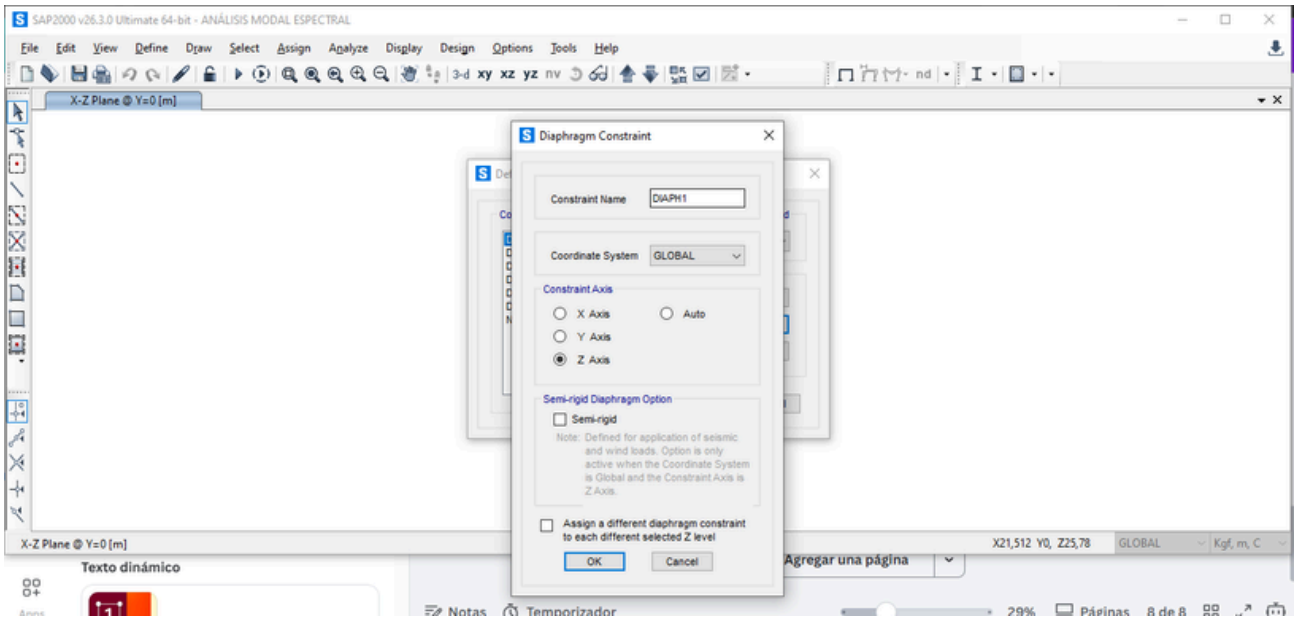
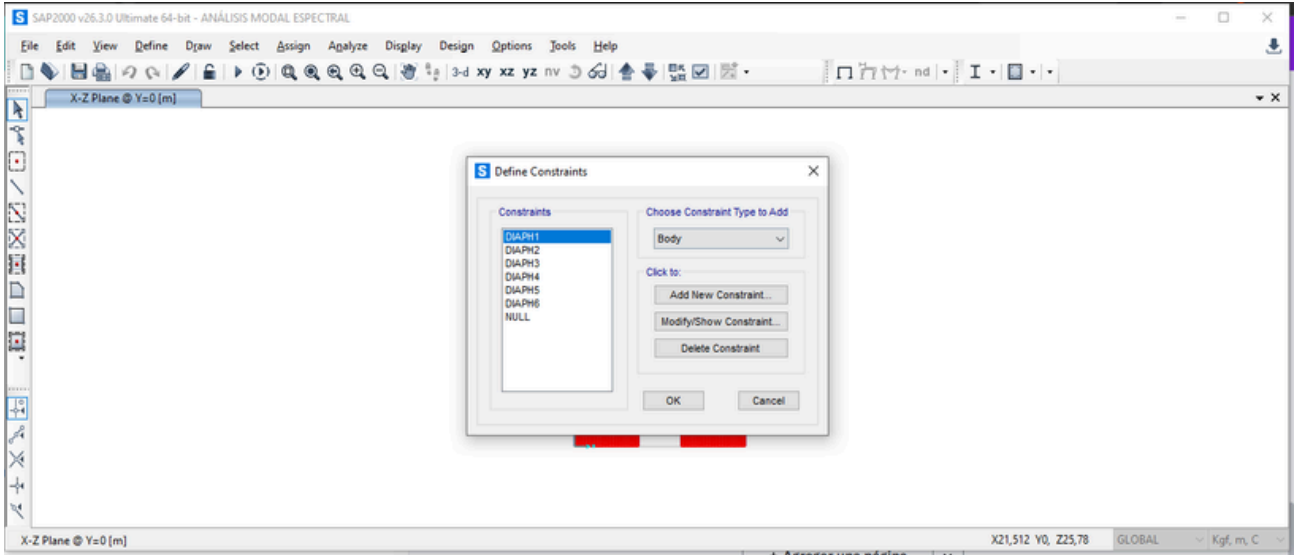


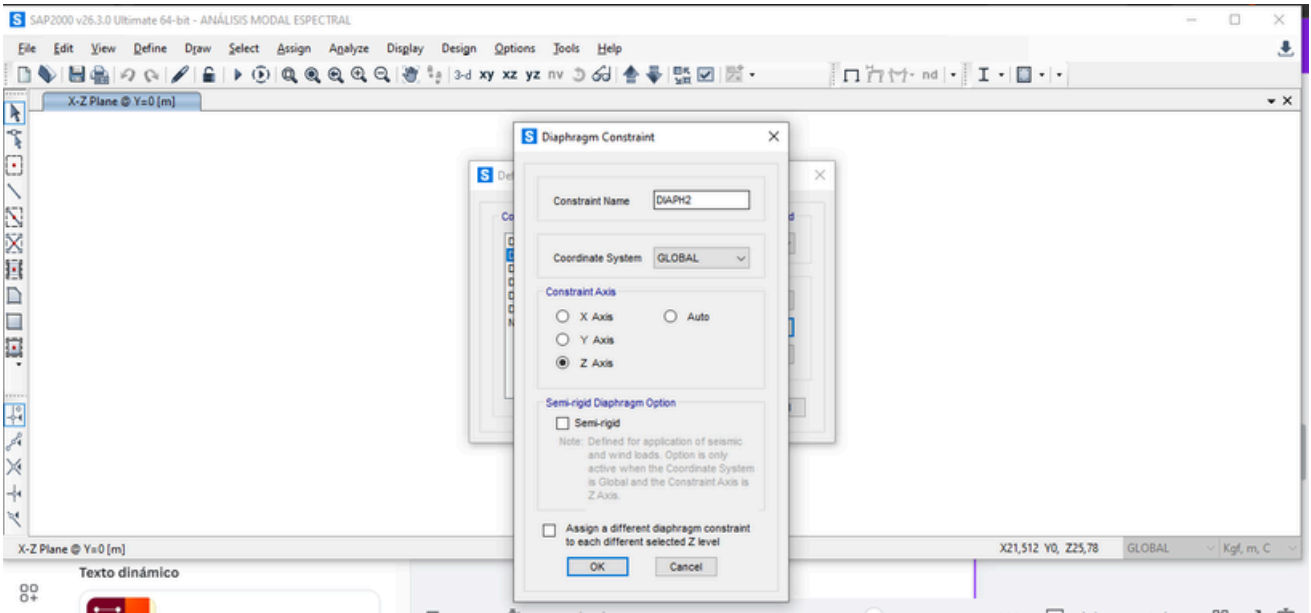


ESTABLECIENDO RESTRICCIONES

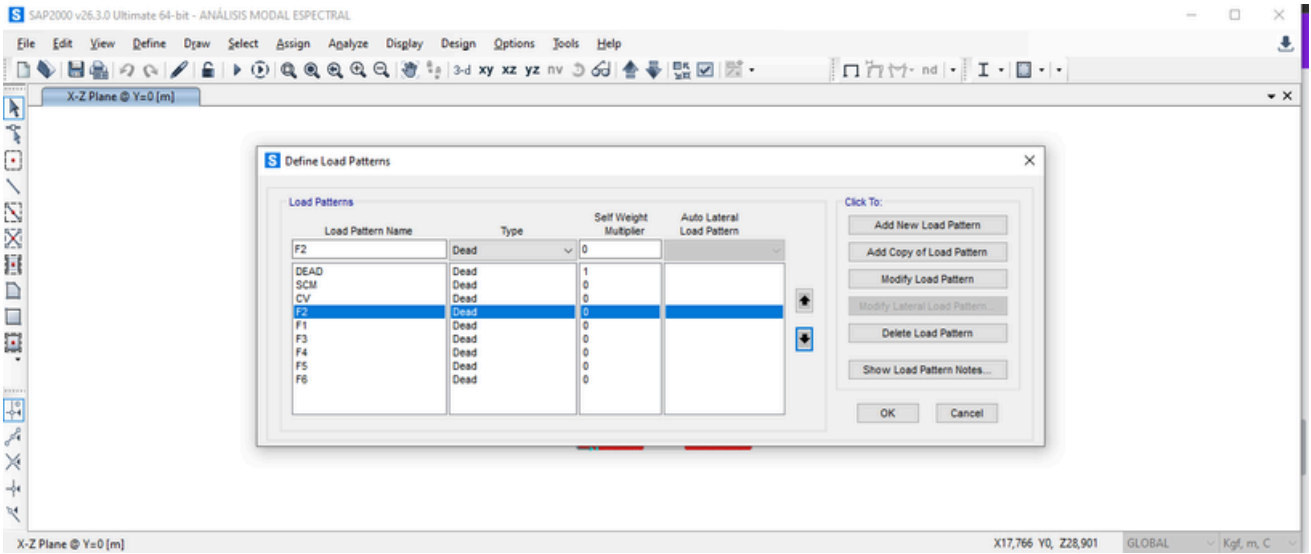


AGREGANDO DIAFRAGMAS RIGIDOS

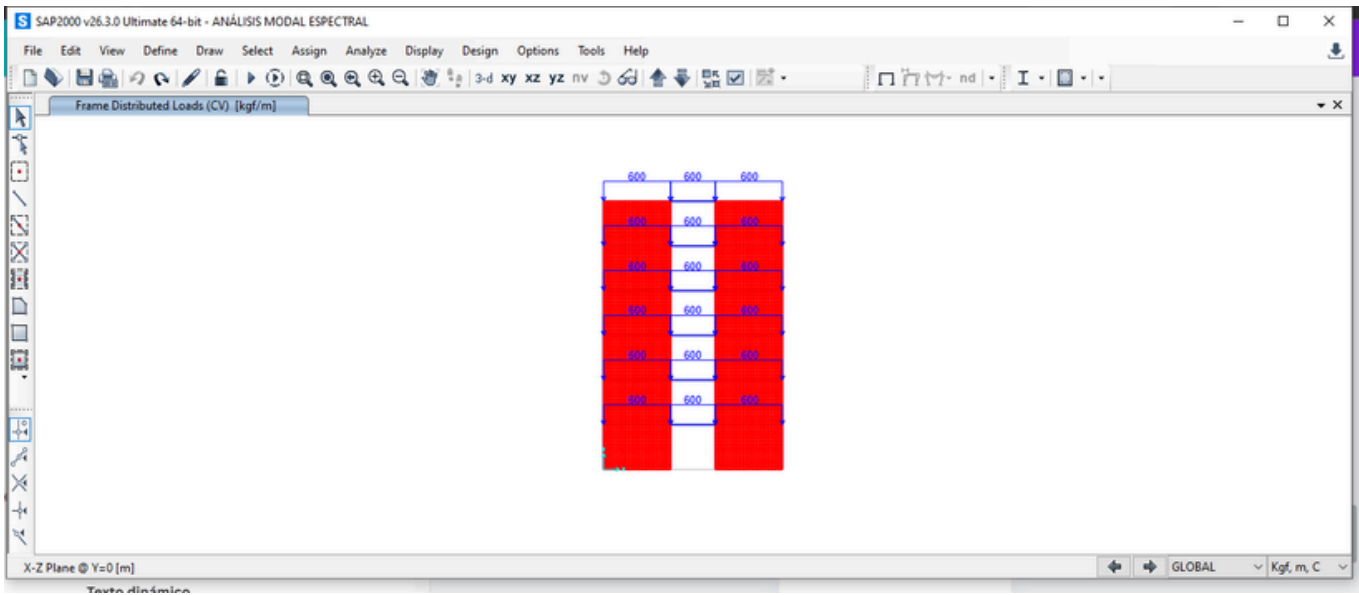
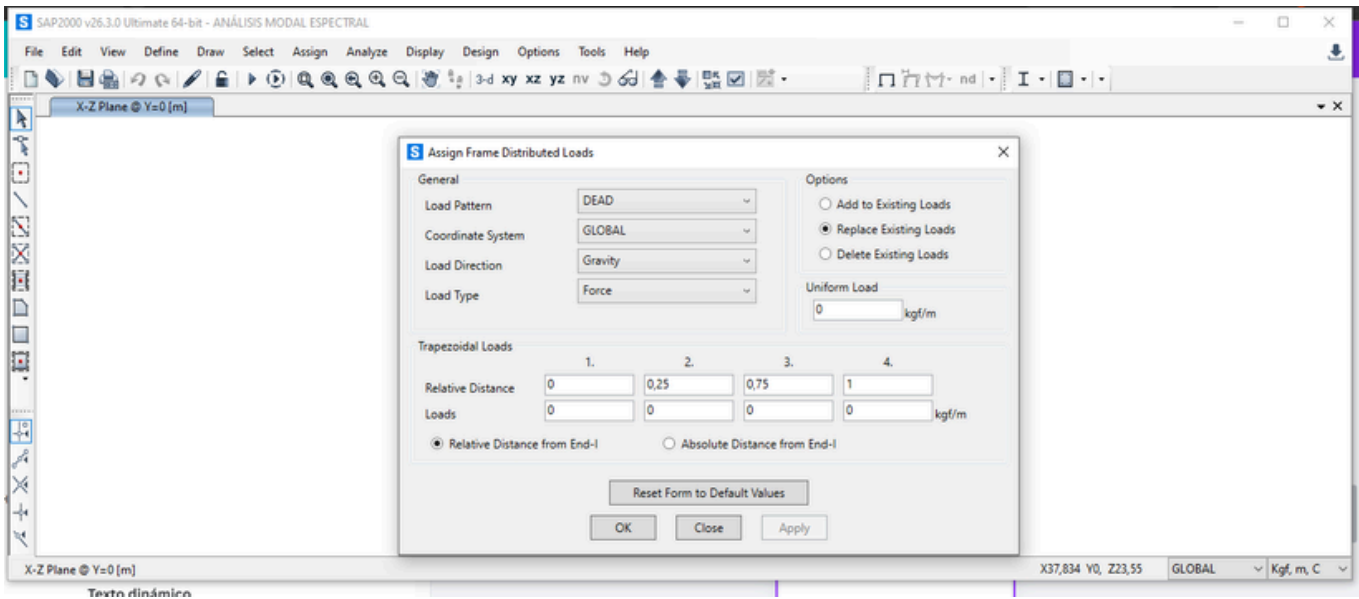
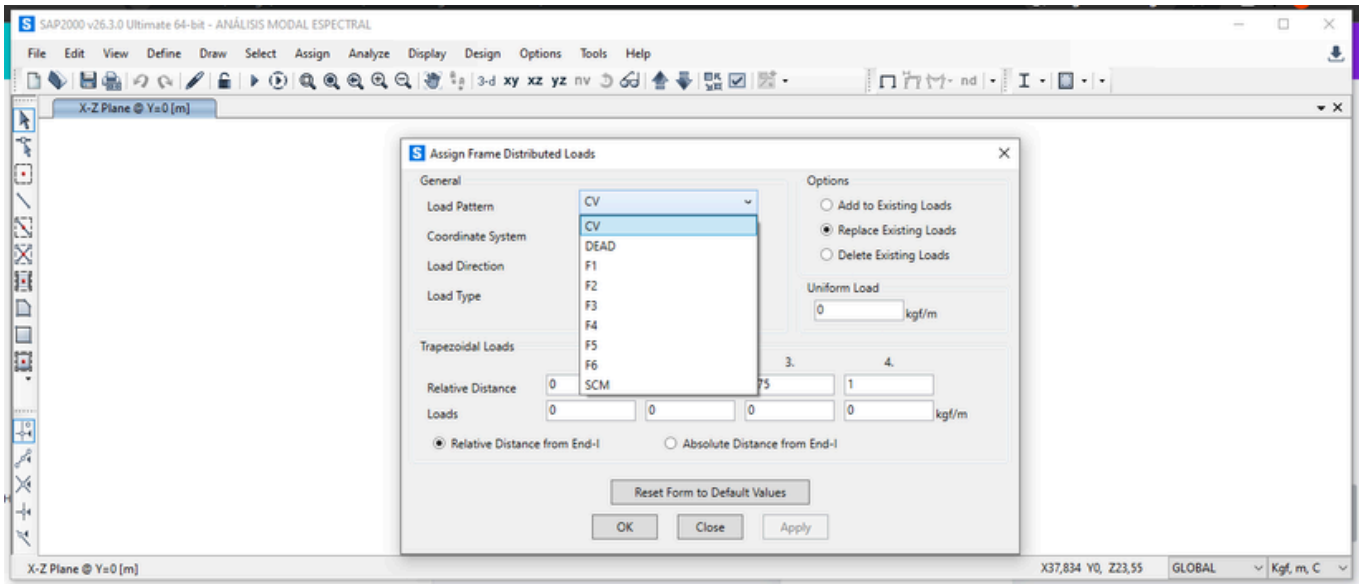




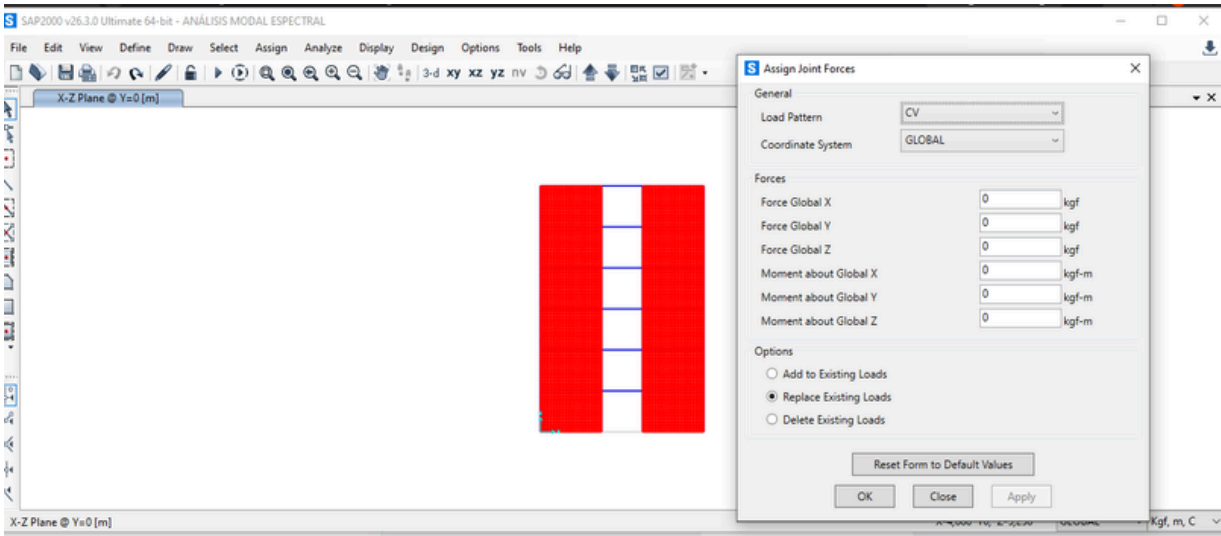
ASIGNANDO CARGAS



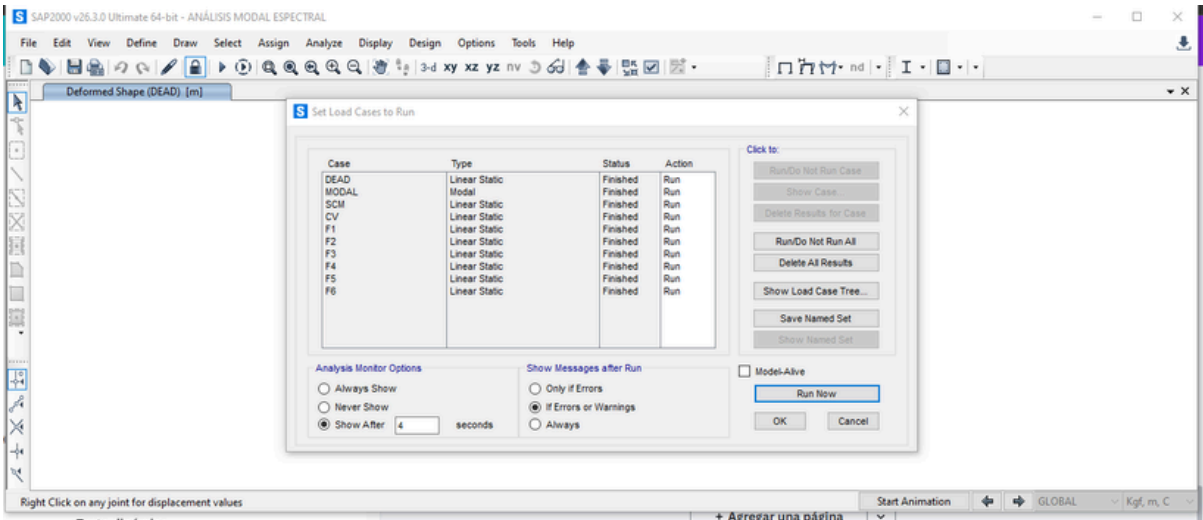
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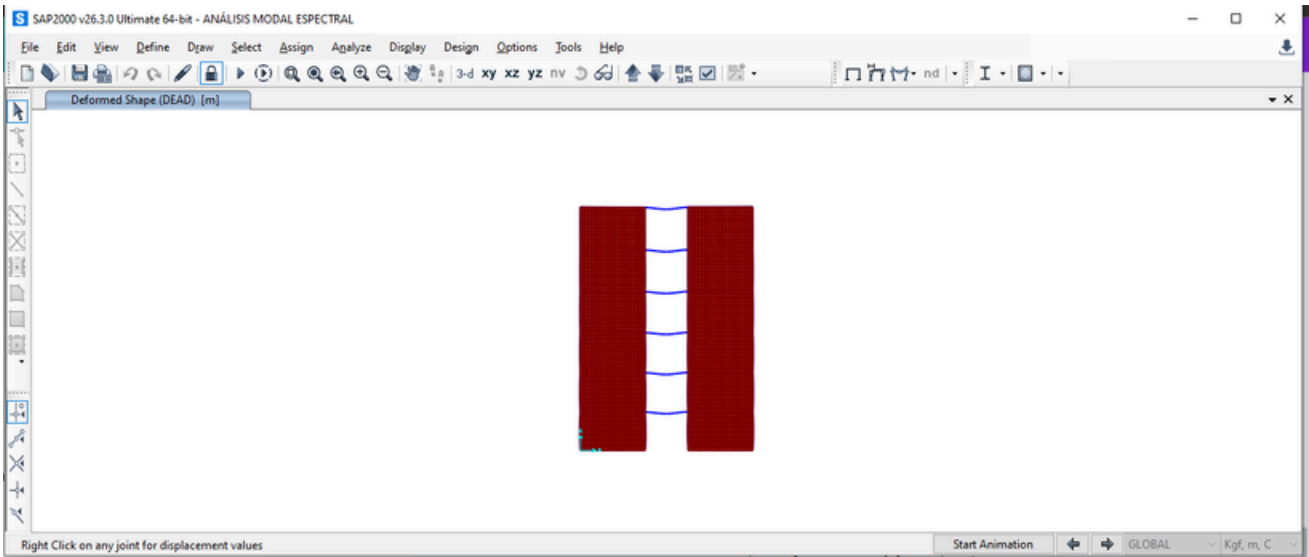


ASIGNANDO FUERZAS EN EL NODO

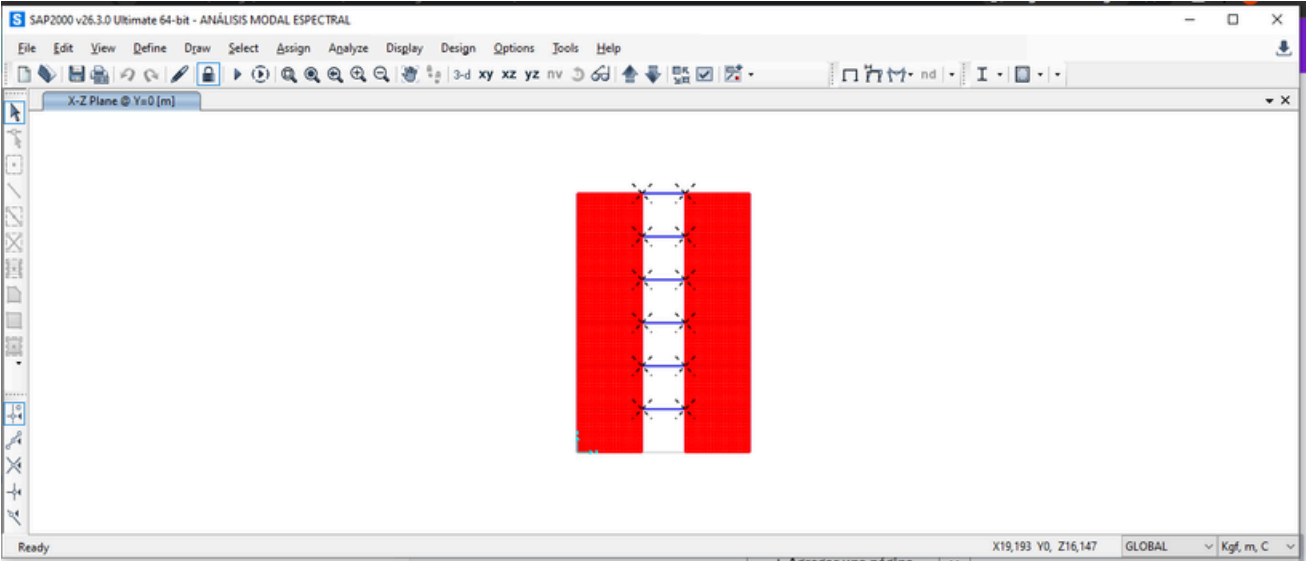


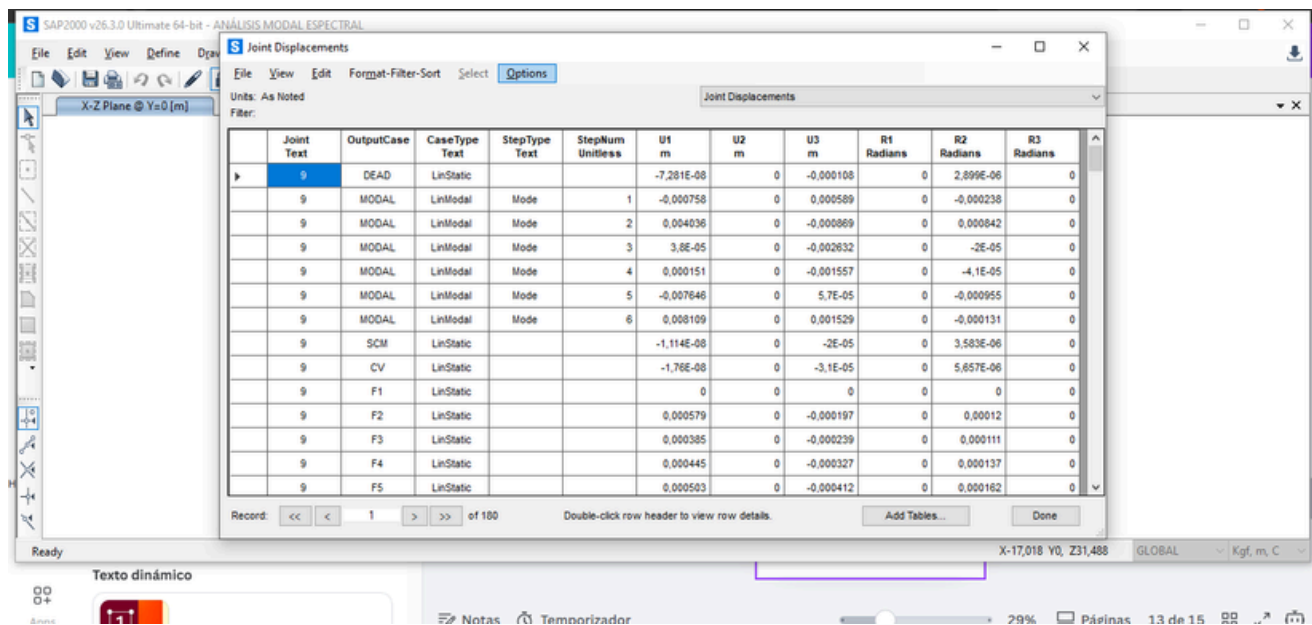
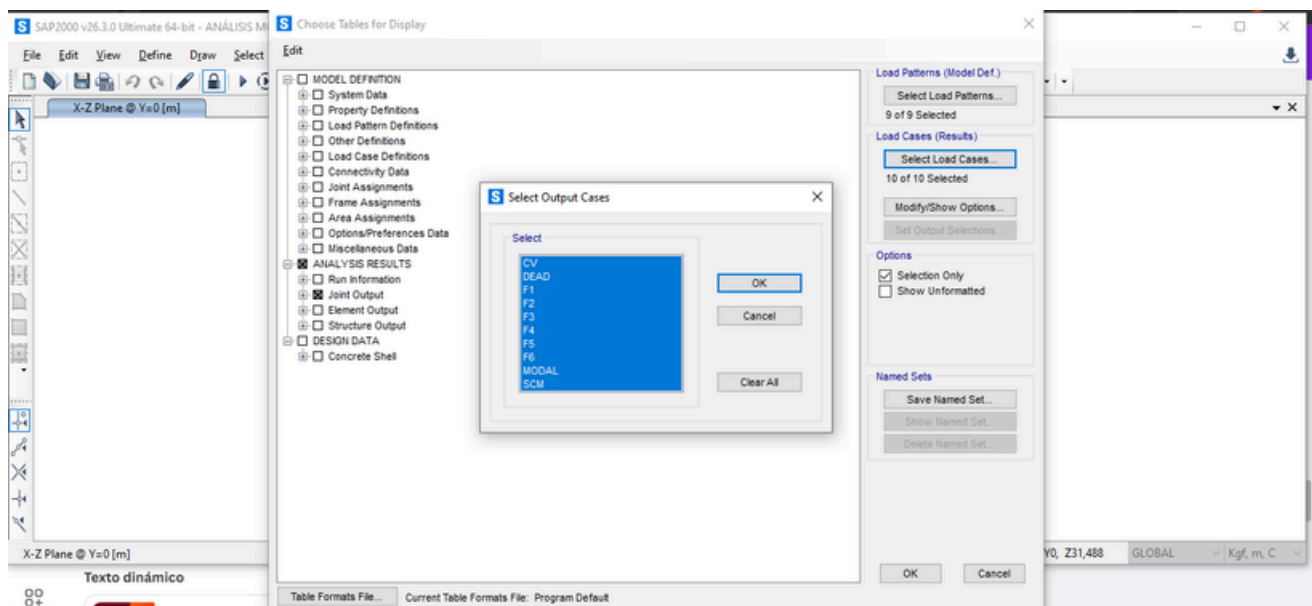
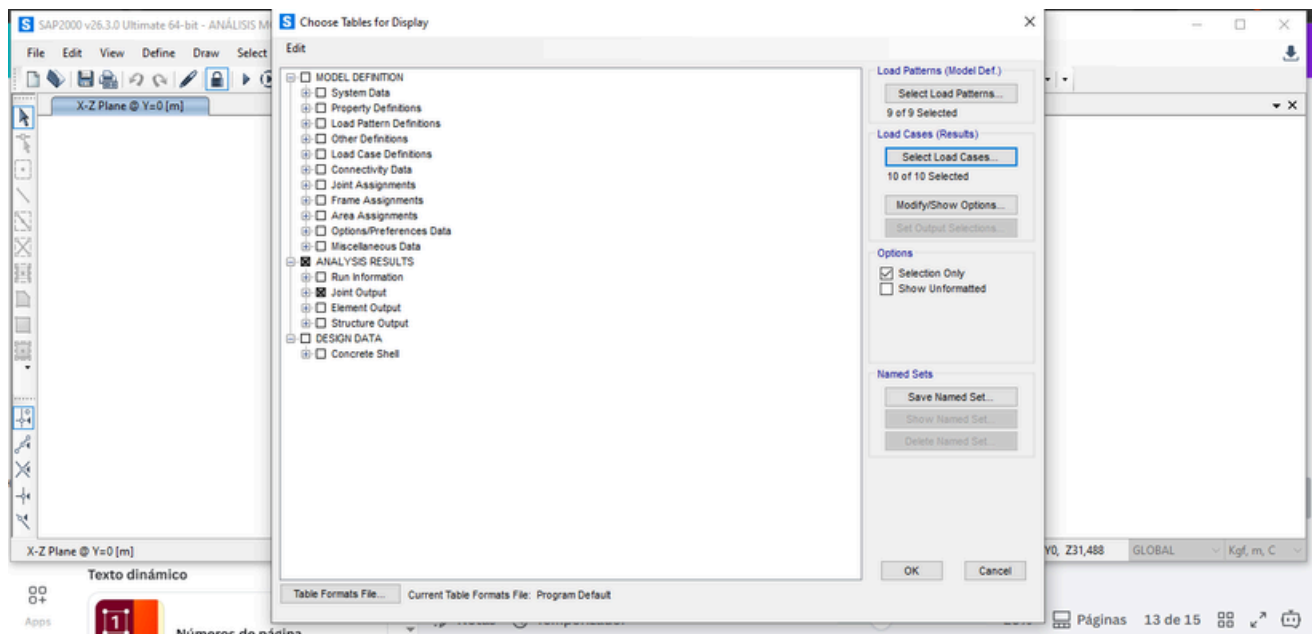
CORRIENDO EL PROGRAMA





OBTENIENDO INFORMACIÓN DE TABLAS





SAP2000 v26.3.0 Ultimate 64-bit - ANÁLISIS MODAL ESPECTRAL

File Edit View Define Draw

File View Edit Format-Filter-Sort Select Options

Units: As Noted

Filter:

Joint Displacements

Joint Text	OutputCase	CaseType Text	StepType Text	StepNum Unitless	U1 m	U2 m	U3 m	R1 Radians	R2 Radians	R3 Radians
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9	MODAL	LinModal	Mode	1	-0.000758	0	0.000589	0	-0.000238	0
9	MODAL	LinModal	Mode	2	0.004036	0	-0.000069	0	0.000842	0
9	MODAL	LinModal	Mode	3	3.8E-05	0	-0.002632	0	-2E-05	0
9	MODAL	LinModal	Mode	4	0.000151	0	-0.001557	0	-4.1E-05	0
9	MODAL	LinModal	Mode	5	-0.007646	0	5.7E-05	0	-0.000955	0
9	MODAL	LinModal	Mode	6	0.008109	0	0.001529	0	-0.000131	0
9	SCM	LinStatic			-1.114E-08	0	-2E-05	0	3.583E-06	0
9	CV	LinStatic			-1.76E-08	0	-3.1E-05	0	5.657E-06	0
9	F1	LinStatic			0	0	0	0	0	0
9	F2	LinStatic			0.000579	0	-0.000197	0	0.00012	0
9	F3	LinStatic			0.000385	0	-0.000239	0	0.000111	0
9	F4	LinStatic			0.000445	0	-0.000327	0	0.000137	0
9	F5	LinStatic			0.000503	0	-0.000412	0	0.000162	0

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Ready X-13,634 Y0, Z30,36 GLOBAL Kgf, m, C

Texto dinámico

SAP2000 v26.3.0 Ultimate 64-bit - ANÁLISIS MODAL ESPECTRAL

File Edit View Define Draw

File View Edit Format-Filter-Sort Select Options

Units: As Noted

Filter:

Joint Displacements

Joint Text	OutputCase	CaseType Text	StepType Text	StepNum Unitless	U1 m	U2 m	U3 m	R1 Radians	R2 Radians	R3 Radians
11	MODAL	LinModal	Mode	3	6.7E-05	0	-0.007191	0	-6.9E-05	0
11	MODAL	LinModal	Mode	4	0.000258	0	-0.00413	0	-0.000215	0
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11	CV	LinStatic			-3.6E-08	0	-7.7E-05	0	5.639E-06	0
11	F1	LinStatic			0	0	0	0	0	0
11	F2	LinStatic			0.001424	0	-0.000221	0	6.5E-05	0
11	F3	LinStatic			0.001814	0	-0.000407	0	0.000142	0
11	F4	LinStatic			0.002377	0	-0.000677	0	0.000244	0
11	F5	LinStatic			0.002924	0	-0.000939	0	0.000322	0
11	F6	LinStatic			0.003463	0	-0.001194	0	0.000399	0
12	DEAD	LinStatic			-8.783E-07	0	-0.000314	0	2.951E-06	0
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Ready X-13,634 Y0, Z30,36 GLOBAL Kgf, m, C